CCNA Day 4 Study Notes

IPv4 Addressing and Subnetting - Part 2

Subnetting - based on the TBN (TungaBadra Networks) Day 4 class handout - CCNA 200-301 Subnetting

1. What is subnetting?

Subnetting is the process of **borrowing bits** to divide one network into multiple smaller networks, based on the company's requirement, and to save IP addresses.

- For subnets (more networks): borrow from the host bits, left to right.
- For hosts (more hosts per network): the requirement decides how many host bits must be kept, counting from right to left.
- Subnets available = 2^(borrowed bits) - you need 2^N >= the number of subnets required.
- Usable hosts per subnet = 2^(host bits) - 2 - you need 2^h - 2 >= the hosts required.
- Advantages: reduces wastage of IP addresses and controls broadcast traffic (each subnet is its own broadcast domain).

2. Worked example - subnetting for company departments

TungaBadra Networks has 3 departments: Tech Team (35 hosts), Sales Team (30 hosts) and HR Team (40 hosts). Give each department its own subnet from 192.168.1.0/24, using one fixed mask for all subnets (FLSM).

Step 1: Networks needed X = 3. Find N with 2^N >= X: 2^2 = 4 >= 3, so borrow N = 2 host bits.

Step 2: New mask: /24 + 2 = /26 -> 255.255.255.192. Each subnet has 64 addresses, 62 usable hosts.

Step 3: Assign: Tech gets 192.168.1.0/26 (62 usable - 35 = 27 free), Sales gets 192.168.1.64/26 (62 - 30 = 32 free), HR gets 192.168.1.128/26 (62 - 40 = 22 free).

Step 4: The 4th block, 192.168.1.192/26, stays spare for a future department.



192.168.1.0/24 borrowed 2 bits -> /26 (each block: 64 addresses, 62 usable hosts)

Class handout correction: this is a fixed-size (FLSM) teaching exercise: every subnet gets the same /26 mask for simplicity, so it is not the most address-efficient design - VLSM would size each subnet to its department (see interview Q5). The handout's extra lines about '192.168.2.0 - 192.168.2.127' do not belong to this worked example (the whole design fits inside 192.168.1.0/24) and have been left out. The /26 plan above is the intended solution.

3. Worked example 1 - 7 subnets from 192.168.0.0/24

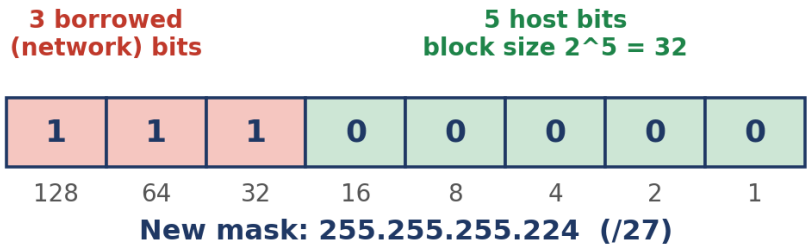
A Class C address 192.168.0.0 needs 7 subnets. Find the new subnet mask and the network addresses.

Step 1: Default Class C mask: 255.255.255.0 (/24).

Step 2: Subnets needed = 7. 2^N >= 7 -> 2^3 = 8 works, so N = 3 borrowed bits. (More subnets than needed is allowed; fewer is not.)

Step 3: New mask: /24 + 3 = /27 -> 255.255.255.224.

Step 4: The 4th octet holds the last network bits. Host bits in the 4th octet = 5, so the adding factor = 2^5 = 32. Count networks by 32 in the 4th octet.



Subnet	Network address
1st	192.168.0.0/27
2nd	192.168.0.32/27
3rd	192.168.0.64/27
4th	192.168.0.96/27
5th	192.168.0.128/27
6th	192.168.0.160/27
7th	192.168.0.192/27
8th	192.168.0.224/27

4. Worked example 2 - 50 subnets from 10.0.0.0/8

Step 1: Default Class A mask: 255.0.0.0 (/8).

Step 2: Networks needed = 50. 2^N >= 50 -> 2^6 = 64 works, so N = 6 borrowed bits.

Step 3: New mask: /8 + 6 = /14 -> 255.252.0.0.

Step 4: The 2nd octet holds the last network bits. Host bits in the 2nd octet = 2, so the adding factor = 2^2 = 4. Count networks by 4 in the 2nd octet.

Step 5: 1st network: 10.0.0.0/14. 2nd: 10.4.0.0/14. 3rd: 10.8.0.0/14 (and so on, up to 10.252.0.0/14).

5. The block-size (interesting octet) method

This is the fastest way to answer exam subnetting questions. Five steps:

Step 1: Identify the interesting octet - the first octet that contains a 0 in the binary mask (the octet where the mask stops being 255).

Step 2: Read the mask's decimal value in that interesting octet.

Step 3: Block size = 256 - that decimal value.

Step 4: Count the subnets in the interesting octet by the block size, starting at 0.

Step 5: Read off the subnet address, broadcast address (one less than the next subnet) and the usable range (everything between them).

Handy reference tables

Host bits h	1	2	3	4	5	6	7	8	9	10
Hosts 2^h - 2	0	2	6	14	30	62	126	254	510	1022

Borrowed bits N	1	2	3	4	5	6	7	8
Subnets 2^N	2	4	8	16	32	64	128	256

Class handout correction: the 2^h - 2 host counts above hold for normal host subnets (up to /30). Exceptions: a /31 point-to-point link uses both addresses - no network or broadcast address (RFC 3021) - and a /32 is a single host route.

6. Practice questions with worked solutions

The 12 exam-style questions from the class handout, each with the block-size method worked through. Cover the answer, solve it yourself, then check.

Question 1. What is the range of assignable IP addresses for the subnet containing 172.16.1.10 /19?

a. 172.16.0.1 - 172.16.31.254

b. 172.16.0.1 - 172.16.63.254

c. 172.16.0.0 - 172.16.31.255

d. 172.16.0.1 - 172.16.31.255

e. 172.16.0.0 - 172.16.63.254

Answer: a

- /19 -> mask 255.255.224.0; interesting octet is the 3rd, value 224, block size 32.
- Subnets: 172.16.0.0, 172.16.32.0, 172.16.64.0 ... 172.16.1.10 sits in 172.16.0.0/19.
- Broadcast is one less than the next subnet: 172.16.31.255. Usable range: 172.16.0.1 - 172.16.31.254.

Question 2. You are assigning IP addresses to hosts in the 192.168.4.0 /26 subnet. Which two of the following IP addresses are assignable? a. 192.168.4.0 b. 192.168.4.63 c. 192.168.4.62 d. 192.168.4.32 e. 192.168.4.64

Answer: c and d

- /26 -> block size 64: subnets .0, .64, .128, .192.
- For 192.168.4.0/26 the usable range is .1 - .62. So .0 (network) and .63 (broadcast) are not assignable; .62 and .32 are.

Question 3. A host has 192.168.181.182 /25. What is the subnet to which the host belongs?

a. 192.168.181.128 /25

b. 192.168.181.0 /25

c. 192.168.181.176 /25

d. 192.168.181.192 /25

e. 192.168.181.160 /25

Answer: a

- /25 -> mask 255.255.255.128; block size 128 in the 4th octet: subnets .0 and .128.
- 182 falls in the .128 - .255 block, so the subnet is 192.168.181.128/25 (usable .129 - .254).

Question 4. You have Class B 172.16.0.0 /16. Maximize the number of broadcast domains where each must hold 1000 hosts. What subnet mask?

a. /22

b. /23

c. /24

d. /25

e. /26

Answer: a

- A subnet = a broadcast domain. Need 2^h - 2 >= 1000 -> h = 10 (1022 hosts).
- 32 - 10 = 22 bits of mask -> /22. Using no more host bits than needed maximizes the number of subnets.

Question 5. What is the directed broadcast address of the subnet containing 172.16.1.10 /19?

a. 172.16.15.255

b. 172.16.31.255

c. 172.16.255.255

d. 172.16.95.255

e. 172.16.0.255

Answer: b

- Same work as Q1: subnet 172.16.0.0/19, next subnet 172.16.32.0, so the directed broadcast is one less: 172.16.31.255.

Question 6. A Class C network 192.168.10.0 is subnetted with a 28-bit mask. How many subnets can be created?

a. 32

b. 16

c. 30

d. 8

e. 14

Answer: b

- Class C natural mask is /24, so /28 means 4 borrowed bits.
- Subnets = 2^4 = 16.

Question 7. Given subnet 172.16.56.0 /21, which two addresses belong to this subnet? a. 172.16.54.129 b. 172.16.62.255 c. 172.16.61.0 d. 172.16.65.255 e. 172.16.64.1

Answer: b and c

- /21 -> block size 8 in the 3rd octet: subnet 172.16.56.0 spans 172.16.56.0 - 172.16.63.255. Usable: 172.16.56.1 - 172.16.63.254.
- 172.16.62.255 and 172.16.61.0 are inside that range. With 11 host bits, a 0 or 255 in the 4th octet does NOT automatically make an address a network or broadcast address - that rule only applies at the subnet's own edges.

Question 8. What is the subnet address of 192.168.5.55 with mask 255.255.255.224?

a. 192.168.5.0 /27

b. 192.168.5.16 /27

c. 192.168.5.32 /27

d. 192.168.5.48 /27

e. 192.168.5.64 /27

Answer: c

- 255.255.255.224 = /27 -> block size 32: subnets .0, .32, .64, ...
- 55 falls in the .32 - .63 block, so the subnet is 192.168.5.32/27.

Question 9. Split 192.168.1.0 /24 for multiple locations. The largest subnet needs 13 hosts. Which mask fits 13 hosts and maximizes the number of subnets?

a. 255.255.255.248

b. 255.255.255.224

c. 255.255.255.252

d. 255.255.255.192

e. 255.255.255.240

Answer: e

- Need 2^h - 2 >= 13 -> h = 4 (14 hosts); h = 3 gives only 6.
- 32 - 4 = 28 bits of mask -> /28 = 255.255.255.240. No extra host bits wasted, so the subnet count stays maximal.

Question 10. A Class C network 192.168.10.0 is subnetted with a 28-bit mask. How many assignable addresses are in each subnet?

a. 32

b. 16

c. 30

d. 8

e. 14

Answer: e

- /28 leaves 32 - 28 = 4 host bits.
- Assignable addresses = 2^4 - 2 = 14.

Question 11. An IP address of 192.168.0.100 /27 belongs to which subnet?

a. 192.168.0.92

b. 192.168.0.128

c. 192.168.0.64

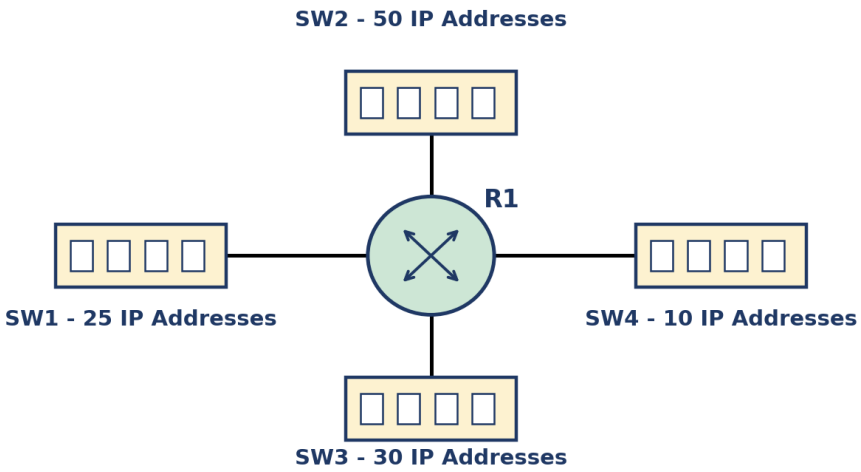
d. 192.168.0.96

e. 192.168.0.32

Answer: d

- /27 -> block size 32: subnets .0, .32, .64, .96, .128, ...
- 100 falls in the .96 - .127 block, so the subnet is 192.168.0.96/27.

Question 12. 192.168.10.0 must support the shown topology: a router with four LANs of 50, 30, 25 and 10 host addresses. Which subnet mask?



a. 255.255.255.0

b. 255.255.255.128

c. 255.255.255.192

d. 255.255.255.224

e. 255.255.255.240

Answer: c

- Four subnets needed -> borrow 2 bits (2^2 = 4). That leaves 6 host bits.
- 2^6 - 2 = 62 hosts per subnet, enough for the largest LAN (50).
- Mask = /26 = 255.255.255.192.

Class handout correction: small typos in the handout solutions (answers themselves are correct): Q7 calls the given subnet '172.16.56.0 /24' (it is /21) and once writes '172.16.54.0' where it means 172.16.56.0; Q11's solution says '19-bit subnet mask' where it means the 27-bit mask from the question.

7. Interview questions and answers

Q1. What is subnetting?

- Based on the requirement, one network is divided into multiple networks by borrowing host bits - this also saves IP addresses.
- If the requirement is networks, bits are borrowed from the host portion (left to right). If the requirement is hosts, enough host bits must be kept (counted right to left).
- Advantages: reduces wastage of IPs and controls broadcast traffic.

Q2. What is the borrowing-bit formula in subnetting?

- 2^N >= X, where X = required networks (subnets) and N = the number of borrowed bits.
- (The handout prints this as '2*N = X' - it is 2 to the power N, and the result must be greater than or equal to what is required.)

Q3. Why do we subtract 2 when counting usable hosts? (added by Instinct - beyond the class handout)

- Two addresses in every subnet are reserved: the network address (all host bits 0) and the broadcast address (all host bits 1).
- So usable hosts = 2^(host bits) - 2. The exceptions are /31 point-to-point links (RFC 3021 - no network or broadcast address) and /32 host routes.

Q4. How do you pick a mask when both the subnet count and the host count matter? (added by Instinct - beyond the class handout)

- First borrow the minimum bits that give enough subnets (2^N >= subnets).
- Then check the remaining host bits give enough hosts (2^h - 2 >= largest LAN).
- If both pass, that mask is the answer - as in practice question 12.

Q5. What is VLSM? (added by Instinct - beyond the class handout)

- Variable Length Subnet Masking: using different subnet masks for different subnets of the same network, so each subnet is sized to its host requirement.
- The examples in this class use one fixed mask (FLSM); VLSM is the natural next step.

Q6. Can a subnet end with a .0 or .255 in the last octet and still be usable? (added by Instinct - beyond the class handout)

- Yes, when there are more than 8 host bits - for example 172.16.61.0 and 172.16.62.255 inside 172.16.56.0/21 are ordinary usable host addresses.
- Only the subnet's own first (all host bits 0) and last (all host bits 1) addresses are reserved.

8. Quick revision sheet

- Subnetting = borrowing host bits to make more networks: 2^N >= subnets needed.
- Hosts per subnet = 2^h - 2 (subtract network + broadcast; exceptions: /31 point-to-point links and /32 host routes - RFC 3021).

- New mask = default mask + borrowed bits. Adding factor = 2^(host bits in the interesting octet).
- Block-size method: interesting octet -> mask value -> 256 minus value = block size -> count subnets -> read off network/broadcast/usable range.
- A subnet is one broadcast domain. 0 or 255 in the last octet is not automatically reserved when host bits > 8.
- Exam speed: memorize the 2^h - 2 table (2, 6, 14, 30, 62, 126, 254, 510, 1022) and the mask chart from Part 1.